

SQL Practice — Physician & Department Database

JOINS, GROUP BY, SUBQUERIES & AGGREGATE FUNCTIONS

Name: Hansika

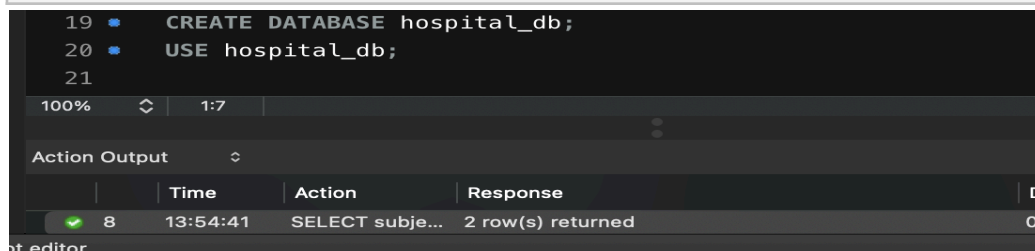
Roll No: 2520030237

Section: 7

Email: gunapatihansikareddy@klh.edu.in

Step 1 — Create Database

```
CREATE DATABASE hospital_db;  
USE hospital_db;
```

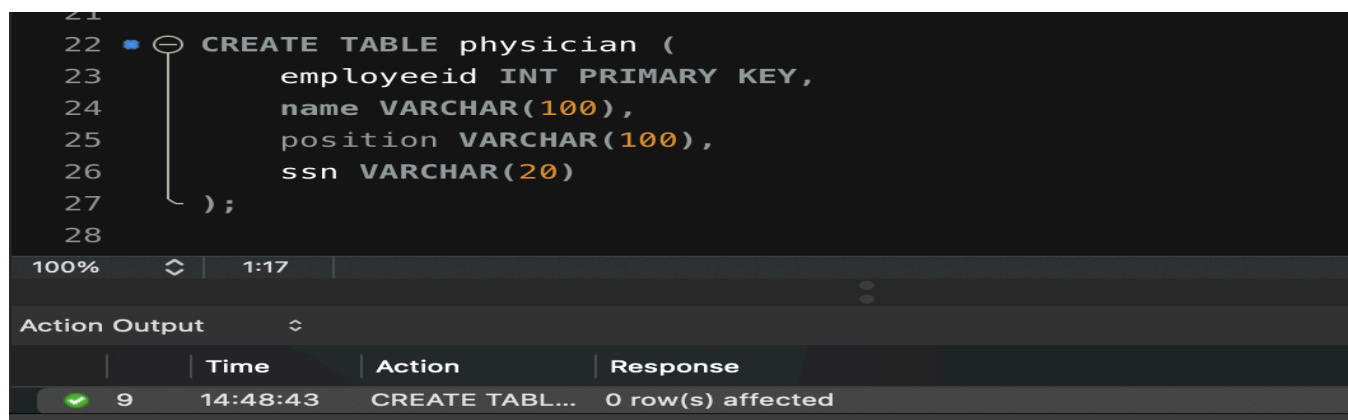


The screenshot shows a SQL IDE with a dark theme. The top pane contains the SQL commands: `19 CREATE DATABASE hospital_db;` and `20 USE hospital_db;`. The bottom pane, titled 'Action Output', shows a table with columns 'Time', 'Action', and 'Response'. A single row is displayed: `13:54:41 SELECT subje... 2 row(s) returned`. The status bar at the bottom indicates '2 row(s) returned'.

	Time	Action	Response
8	13:54:41	SELECT subje...	2 row(s) returned

Step 2 — Create Physician Table (DDL)

```
CREATE TABLE physician (  
    employeeid INT PRIMARY KEY,  
    name VARCHAR(100),  
    position VARCHAR(100),  
    ssn VARCHAR(20)  
);
```



The screenshot shows a SQL IDE with a dark theme. The top pane contains the SQL command: `22 CREATE TABLE physician (
23 employeeid INT PRIMARY KEY,
24 name VARCHAR(100),
25 position VARCHAR(100),
26 ssn VARCHAR(20)
27);`. The bottom pane, titled 'Action Output', shows a table with columns 'Time', 'Action', and 'Response'. A single row is displayed: `14:48:43 CREATE TABL... 0 row(s) affected`. The status bar at the bottom indicates '0 row(s) affected'.

	Time	Action	Response
9	14:48:43	CREATE TABL...	0 row(s) affected

Step 3 — Insert Data into Physician and Display

```
INSERT INTO physician VALUES
(1,'Hansika Reddy','Staff Internist','11111111'),
(2,'Priya Sharma','Attending Physician','22222222'),
(3,'Arjun Iyer','Surgical Attending Physician','33333333'),
(4,'Aditya Rao','Senior Attending Physician','44444444'),
(5,'Rohan Mehta','Head Chief of Medicine','55555555'),
(6,'Vikram Nair','Surgical Attending Physician','66666666'),
(7,'Aditya Verma','Surgical Attending Physician','77777777'),
(8,'Karthik Iyer','MD Resident','88888888'),
(9,'Sneha Pillai','Attending Psychiatrist','99999999');

SELECT * FROM physician;
```

	employeeid	name	position	ssn	
	1	Hansika Reddy	Staff Internist	11111111	
	2	Priya Sharma	Attending Physician	22222222	
	3	Arjun Iyer	Surgical Attending Physician	33333333	
	4	Aditya Rao	Senior Attending Physician	44444444	
	5	Rohan Mehta	Head Chief of Medicine	55555555	
	6	Vikram Nair	Surgical Attending Physician	66666666	
	7	Aditya Verma	Surgical Attending Physician	77777777	
	8	Karthik Iyer	MD Resident	88888888	
	9	Sneha Pillai	Attending Psychiatrist	99999999	
	NULL	NULL	NULL	NULL	

Step 4 — Create Department Table (DDL)

```
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```

40	•	⊖	CREATE TABLE department (
41			departmentid INT PRIMARY KEY,
42			name VARCHAR(100),
43			head INT
44			);
45			
100% 1:35			
Action Output			
		Time	Action
			Response
			Du
✓	12	14:56:46	CREATE TABL...
			0 row(s) affected
			0.0

Step 5 — Insert Data into Department and Display

```
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
```

```
SELECT * FROM department;
```

1	General Medicine	4	
2	Surgery	7	
3	Psychiatry	9	
NULL	NULL	NULL	

Step 6 — Question 1: INNER JOIN

```
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
```

Department	Head_Physician	
General Medicine	Aditya Rao	
Surgery	Aditya Verma	
Psychiatry	Sneha Pillai	

Step 7 — Question 2: WHERE Clause

```
SELECT *
FROM physician
WHERE position='Surgical Attending Physician';
```

employeeid	name	position	ssn	
3	Arjun Iyer	Surgical Attending Physician	333333333	
6	Vikram Nair	Surgical Attending Physician	666666666	
7	Aditya Verma	Surgical Attending Physician	777777777	
NULL	NULL	NULL	NULL	

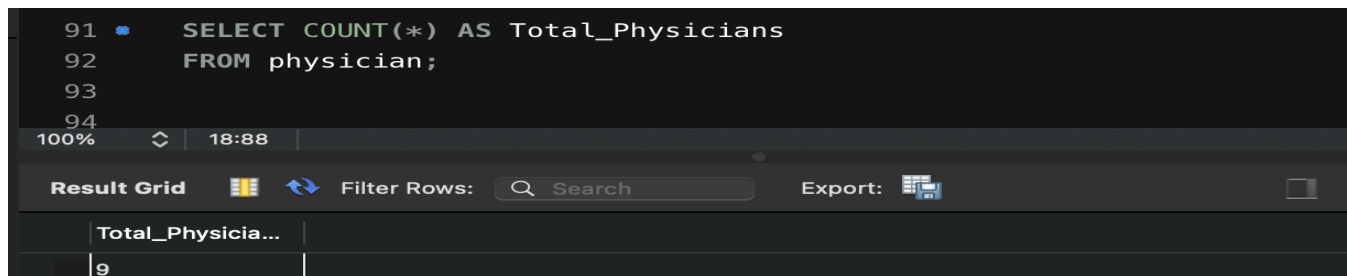
Step 8 — Question 3: WHERE + LIKE

```
SELECT *  
FROM physician  
WHERE name LIKE 'Aditya%';
```

	employeeid	name	position	ssn	
	4	Aditya Rao	Senior Attending Physician	444444444	
	7	Aditya Verma	Surgical Attending Physician	777777777	
	NULL	NULL	NULL	NULL	

Step 9 — Question 4: COUNT()

```
SELECT COUNT(*) AS Total_Physicians  
FROM physician;
```



```
91 SELECT COUNT(*) AS Total_Physicians  
92 FROM physician;  
93  
94
```

100% 18:88

Result Grid Filter Rows: Search Export:

Total_Physicia...
9

Step 10 — Question 5: COUNT DISTINCT

```
SELECT COUNT(DISTINCT position)  
FROM physician;
```

COUNT(DISTINCT positio...
7

Step 11 — Question 6: GROUP BY

```
SELECT position,  
       COUNT(*) AS Total_Employees  
FROM physician  
GROUP BY position;
```

	position	Total_Employees	
	Staff Internist	1	
	Attending Physician	1	
	Surgical Attending Physician	3	
	Senior Attending Physician	1	
	Head Chief of Medicine	1	
	MD Resident	1	
	Attending Psychiatrist	1	

Step 12 — Question 7: GROUP BY + HAVING

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

	position	Total_Employees	
	Surgical Attending Physician	3	

Step 13 — Question 8: ORDER BY

```
SELECT *
FROM physician
ORDER BY name ASC;
```

	employeeid	name	position	ssn	
	4	Adity...	Senior Attending Physician	444444444	
	7	Adity...	Surgical Attending Physician	777777777	
	3	Arjun...	Surgical Attending Physician	333333333	
	1	Hans...	Staff Internist	111111111	
	8	Karth...	MD Resident	888888888	
	2	Priya...	Attending Physician	222222222	
	5	Roha...	Head Chief of Medicine	555555555	
	9	Sneh...	Attending Psychiatrist	999999999	
	6	Vikra...	Surgical Attending Physician	666666666	
	NULL	NULL	NULL	NULL	

Step 14 — Question 9: ORDER BY DESC

```
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

	employeeid	name	position	ssn	
	9	Sneh...	Attending Psychiatrist	999999999	
	8	Karth...	MD Resident	888888888	
	7	Adity...	Surgical Attending Physician	777777777	
	6	Vikra...	Surgical Attending Physician	666666666	
	5	Roha...	Head Chief of Medicine	555555555	
	4	Adity...	Senior Attending Physician	444444444	
	3	Arjun...	Surgical Attending Physician	333333333	
	2	Priya...	Attending Physician	222222222	
	1	Hans...	Staff Internist	111111111	
	NULL	NULL	NULL	NULL	

Step 15 — Question 10: WHERE + IN.

```

SELECT *
FROM physician
WHERE employeeid IN
(
    SELECT head
    FROM department
);

```

	employeeid	name	position	ssn	
	4	Adity...	Senior Attending Physician	444444444	
	7	Adity...	Surgical Attending Physician	777777777	
	9	Sneh...	Attending Psychiatrist	999999999	
	NULL	NULL	NULL	NULL	

Step 16 — Question 11: Subquery (NOT IN)

```

SELECT *
FROM physician
WHERE employeeid NOT IN
(
    SELECT head
    FROM department
);

```

	employeeid	name	position	ssn	
	1	Hans...	Staff Internist	111111111	
	2	Priya...	Attending Physician	222222222	
	3	Arjun...	Surgical Attending Physician	333333333	
	5	Roha...	Head Chief of Medicine	555555555	
	6	Vikra...	Surgical Attending Physician	666666666	
	8	Karth...	MD Resident	888888888	
	NULL	NULL	NULL	NULL	

Step 17 — Question 12: GROUP BY + ORDER BY

```
SELECT position,  
       COUNT(*) AS Total_Employees  
FROM physician  
GROUP BY position  
ORDER BY Total_Employees DESC;
```

	position	Total_Employees	
	Surgical Attending Physician	3	
	Staff Internist	1	
	Attending Physician	1	
	Senior Attending Physician	1	
	Head Chief of Medicine	1	
	MD Resident	1	
	Attending Psychiatrist	1	

Step 18 — Question 13: WHERE + LIKE

```
SELECT *  
FROM physician  
WHERE position LIKE '%Attending%';
```

	employeeid	name	position	ssn	
	2	Priya...	Attending Physician	222222222	
	3	Arjun...	Surgical Attending Physician	333333333	
	4	Adity...	Senior Attending Physician	444444444	
	6	Vikra...	Surgical Attending Physician	666666666	
	7	Adity...	Surgical Attending Physician	777777777	
	9	Sneh...	Attending Psychiatrist	999999999	
	NULL	NULL	NULL	NULL	

Step 19 — Question 14: JOIN + WHERE

```
SELECT p.name  
FROM physician p  
JOIN department d  
ON p.employeeid=d.head  
WHERE d.name='Surgery';
```

	name	
	Adity...	

Step 20 — Question 15: JOIN + ORDER BY

```
SELECT d.name,  
       p.name  
FROM department d  
JOIN physician p  
ON d.head=p.employeeid  
ORDER BY d.name;
```

	name	name	
	General Medicine	Adity...	
	Psychiatry	Sneh...	
	Surgery	Adity...	

Step 21 — Question 16: WHERE + NOT LIKE

```
SELECT *  
FROM physician  
WHERE position NOT LIKE '%Surgical%';
```

	employeeid	name	position	ssn	
	1	Hans...	Staff Internist	111111111	
	2	Priya...	Attending Physician	222222222	
	4	Adity...	Senior Attending Physician	444444444	
	5	Roha...	Head Chief of Medicine	555555555	
	8	Karth...	MD Resident	888888888	
	9	Sneh...	Attending Psychiatrist	999999999	
	NULL	NULL	NULL	NULL	

Step 22 — Question 17: COUNT + HAVING

```
SELECT position,  
       COUNT(*) AS Total  
FROM physician  
GROUP BY position  
HAVING COUNT(*) >= 2;
```


Result Grid   Filter Rows: Search Export: 			
	position	Total	
	Surgical Attending Physician	3	

Step 23 — Question 18: EXISTS.

```
SELECT *
FROM department d
WHERE EXISTS
(
  SELECT 1
  FROM physician p
  WHERE p.employeeid=d.head
);
```

	department...	name	head	
	1	General Medicine	4	
	2	Surgery	7	
	3	Psychiatry	9	
	NULL	NULL	NULL	

Step 24 — Question 19: MAX()

```
SELECT MAX(employeeid)
FROM physician;
```

	MAX(employeei...	
	9	

Step 25 — Question 20: MIN()

```
SELECT MIN(employeeid)
FROM physician;
```

	MIN(employeei...	
	1	